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A Groupwise Multilinear Correspondence Optimization for 3D Faces

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Abstract

Multilinear face models are widely used to model the space of human faces with expressions. For databases of 3D human faces of different identities performing multiple expressions, these statistical shape models decouple identity and expression variations. To compute a high-quality multilinear face model, the quality of the registration of the database of 3D face scans used for training is essential. Meanwhile, a multilinear face model can be used as an effective prior to register 3D face scans, which are typically noisy and incomplete. Inspired by the minimum description length approach, we propose the first method to jointly optimize a multilinear model and the registration of the 3D scans used for training. Given an initial registration, our approach fully automatically improves the registration by optimizing an objective function that measures the compactness of the multilinear model, resulting in a sparse model. We choose a continuous representation for each face shape that allows to use a quasi-Newton method in parameter space for optimization. We show that our approach is computationally significantly more efficient and leads to correspondences of higher quality than existing methods based on linear statistical models. This allows us to evaluate our approach on large standard 3D face databases and in the presence of noisy initializations.

1. Introduction

The human face is one important factor for any kind of social interaction in our daily life. This motivates many different fields such as human computer interaction, medicine, ergonomics or security, to investigate the human face. Since many of these areas are interested in the 3D geometry of the face, the number of publicly available 3D face databases increased over the last years. As the manual analysis of large databases is intractable, automatic data driven and statistical approaches are widely used to analyze the structure of the data. To compute statistics, all shapes of the dataset need to be in correspondence [10, Chapter 1].

Computing these correspondences for human face data is a challenging task that many methods aim to solve (*e.g.* [24, 26, 17, 14, 27]). Given a good registration, a statistical face model can be learned. In computer vision and graphics, statistical face models are used *e.g.* to reconstruct the 3D geometry of the face from 2D images [1], to recognize facial expressions [24], to transfer expressions between images or videos [30], or to change expressions in 3D videos [31].

Statistical face models can also be used to reconstruct the 3D geometry from noisy or partially occluded face scans [5] and are therefore directly applicable for registration. Furthermore, registration methods with prior learned knowledge outperform model-free methods like template fitting [25, 2]. Summing up, this is a chicken-and-egg problem: given a good registration, a statistical model can be learned, and given a representative statistical model, a good registration can be computed. The quality of a given statistical model can be measured [10, Chapter 3.3.1], and due to the dependency of the statistical model on a registration, this measurement also evaluates the underlying registration.

Methods that aim at jointly optimizing the registration and a statistical model have been developed for principal component analysis (PCA) (*e.g.* [10, Chapter 4], [21]). Furthermore, variants of this linear method like part-based PCA [6], kernel PCA [8] or human body specific approaches [16] exist. These methods measure the model quality and change the registration such that the quality of the model and the registration improve at the same time. Since the model quality depends on all shapes, these methods are called groupwise optimization methods. Linear PCA-based methods have been proven to outperform different pairwise correspondence optimization methods [9].

Since the variations in databases of human faces from different identities performing different expressions cannot be modeled well using a linear space, the existing methods are not suitable for optimizing the correspondence of human faces. The space of human faces in various expressions can be well modeled using a multilinear model [30, 24, 31, 2, 5], which is a higher-order generalization of a PCA model.

This motivates us to propose an approach to optimize the correspondence for 3D face databases based on multilinear statistical models. The correspondence is optimized

三维人脸的分组多线性对应优化

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摘要

Multilinear face models are widely used to model the space of human faces with expressions. For databases of 3D human faces of different identities performing multiple expressions, these statistical shape models decouple identity and expression variations. To compute a high-quality multilinear face model, the quality of the registration of the database of 3D face scans used for training is essential. Meanwhile, a multilinear face model can be used as an effective prior to register 3D face scans, which are typically noisy and incomplete. Inspired by the minimum description length approach, we propose the first method to jointly optimize a multilinear model and the registration of the 3D scans used for training. Given an initial registration, our approach fully automatically improves the registration by optimizing an objective function that measures the compactness of the multilinear model, resulting in a sparse model. We choose a continuous representation for each face shape that allows to use a quasi-Newton method in parameter space for optimization. We show that our approach is computationally significantly more efficient and leads to correspondences of higher quality than existing methods based on linear statistical models. This allows us to evaluate our approach on large standard 3D face databases and in the presence of noisy initializations.

1. 引言

人脸是我们日常社交互动中的一个重要因素。这推动了许多不同领域，如人机交互、医学、工效学或安全领域，对人脸进行研究。由于这些领域中有许多对脸部的三维几何形状感兴趣，因此近年来公开可用的三维人脸数据库数量有所增加。由于对大型数据库进行手动分析难以处理，自动化的数据驱动和统计方法被广泛用于分析数据结构。为了计算统计数据，数据集中的所有形状都需要建立对应关系[10，第1章]。

计算人脸数据的这些对应关系是许多方法旨在解决的一项具有挑战性的任务（*e.g*。[24, 26, 17, 14, 27]）。在良好配准的情况下，可以学习统计人脸模型。在计算机视觉和图形学中，统计人脸模型被用于*e.g*。从2D图像重建面部3D几何结构[1]、识别面部表情[24]、在图像或视频之间转移表情[30]，或在3D视频中改变表情[31]。

统计面部模型也可用于从含噪或部分遮挡的面部扫描中重建3D几何结构[5]，因此可直接应用于配准。此外，具有先验知识的配准方法优于无模型方法，如模板拟合[25, 2]。综上所述，这是一个“鸡生蛋”问题：给定良好的配准，可以学习统计模型；给定具有代表性的统计模型，可以计算出良好的配准。给定统计模型的质量可以衡量[10, 第3.3.1节]，并且由于统计模型依赖于配准，该衡量也评估了底层配准的质量。

旨在联合优化配准与统计模型的方法已针对主成分分析（PCA）开发（*e.g*。[10，第4章]，[21]）。此外，该线性方法的变体，如基于部分的PCA [6]、核PCA [8]或特定人体方法[16]也存在。这些方法衡量模型质量并调整配准，使模型质量和配准同时得到提升。由于模型质量依赖于所有形状，这些方法被称为组群优化方法。基于线性PCA的方法已被证明优于不同的成对对应优化方法[9]。

由于不同身份的人脸在不同表情下的数据库变化无法用线性空间很好地建模，现有方法不适用于优化人脸的对应关系。利用多线性模型[30, 24, 31, 2, 5]可以很好地建模各种表情下的人脸空间，该模型是PCA模型的高阶推广。

这促使我们提出一种基于多线性统计模型优化3D人脸数据库对应关系的方法。对应关系的优化

based on the minimum description length (MDL) principle, which leads to a sparse multilinear model. A key advantage of extending MDL to multilinear models is a reduced parameter space, which can be optimized efficiently. The main challenge is that while for the linear case PCA provides an optimal low-dimensional space, the solution for the multilinear case is NP-hard to compute [15]. To find a good basis for face models, we compare different tensor decompositions for their ability to reconstruct unseen face data. Another previously unaddressed challenge related to 3D face data is to allow for manifold boundaries during optimization, which is needed as the face has a mouth and an outer boundary. We solve this issue efficiently by introducing constraints in the optimization framework.

The main contributions of this work are: (1) we introduce the first fully automatic groupwise correspondence optimization approach for multilinearly distributed data, and (2) we show that our approach is computationally significantly more efficient and leads to correspondences of higher quality than existing PCA-based optimization methods.

2. Related work

Template-based facial correspondence computation: Our method is related to methods that aim to compute correspondences between sets of shapes. While many methods exist to establish correspondence for arbitrary classes of shapes, we focus on 3D face registration methods. Given a sparse set of 3D landmarks, Mpiperis *et al*. [24] register 3D faces in various expressions with an elastically deformable face model. Passalis *et al*. [26] fit an annotated face model to the scan by solving a second order differential equation. Huang *et al*. [17] split the face into multiple parts and perform a deformation of each part to fit an input face. Guo *et al*. [14] use a thin-plate spline guided template fitting to register 3D face scans. Pan *et al*. [25] use a sparse deformable model for registration. They learn a dictionary on a set of registered faces and register a new face by restricting the correspondences to be a sparse representation of the learned dictionary. Salazar *et al*. [27] use a blendshape model to fit the expression followed by a template fitting using a non-rigid iterative closest point method to get the facial details.

All of these methods find a good correspondence, but none of them aim at producing a registration that is optimal for statistical modeling. Note that any of these methods can be used to initialize our optimization approach.

Statistical face models: Given a set of 3D shapes in full correspondence, various methods can perform statistical analysis. We focus our discussion on multilinear shape spaces for 3D faces. Vlasic *et al*. [30] use a multilinear model for a database of human faces that decouples facial shape, expression, and viseme to transfer facial performance between 2D videos. Mpiperis *et al*. [24] use a multilinear model for identity and expression recognition. Yang *et al*. [31] reconstruct the 3D face shape from 2D videos and exploit the decoupling of identity and expression variations to modify the identity or expression within the videos. Bolkart and Wuhrer [2] use a multilinear model to register a large database of 3D faces in motion and perform analysis on the resulting registration. Brunton *et al*. [5] learn multiple localized multilinear models and use these to reconstruct models from noisy and occluded face scans.

As these methods use a multilinear model for 3D faces of multiple identities and expressions, they employ the same model as our method. However, none of them aim at optimizing the correspondence using the learned model.

Registration optimization: While some prior works in machine learning explore the idea of jointly learning a model and correspondence information (*e.g*. [4, 29, 18]), our method is most related to methods that aim to jointly optimize the registration of a set of 3D shapes and a learned statistical model. Kotcheff and Taylor [21] propose a groupwise correspondence optimization based on a PCA model that explicitly favors compact models. Davies *et al*. [10, Chapter 4] give an overview of different objective functions for correspondence optimization and motivate an information theoretic objective function minimizing the description length of the data. The basic concept of minimum description length approaches is to minimize the length of a message that is transmitted from a sender to a receiver. They encode the data with a PCA model and alter the correspondence such that the number of bits needed to describe the model and the encoded data is minimal. Davies *et al*. [9] show that MDL outperforms state-of-the-art registration methods for medical datasets. Gollmer *et al*. [13] compare different objective functions. They show that while the determinant of the covariance matrix is easier to optimize, the results are comparable to results produced by MDL.

All these methods model the data with one linear PCA model. In contrast, Burghard *et al*. [6] use a part-based linear model, and Chen *et al*. [8] model the data with a non-linear kernel PCA. Hirshberg *et al*. [16] derive a skeleton based approach specifically for human body shapes to jointly optimize the registration and a statistical model.

None of these methods can model 3D faces with varying identities and expressions. To allow this, we introduce the first groupwise correspondence optimization approach for multilinearly distributed data. Furthermore, while most methods assume the object to be a closed manifold, our approach handles manifolds with multiple boundaries.

3. Multilinear shape space

This section introduces the multilinear model and different tensor decompositions to derive the model. Multilinear models can effectively model statistical variations of faces due to identity and expression as it decouples these two types of shape variation.

基于最小描述长度（MDL）原理，该原理引出一个稀疏多线性模型。将MDL扩展至多线性模型的一个关键优势在于参数空间缩减，从而可高效优化。主要挑战在于，线性情形下PCA提供了最优低维空间，而多线性情形的解在计算上是NP难的[15]。为找到人脸模型的良好基底，我们比较了不同张量分解在重建未见人脸数据方面的能力。另一个先前未解决的与3D人脸数据相关的挑战是在优化过程中允许流形边界，这一点是必要的，因为人脸具有嘴巴和外部边界。我们通过在优化框架中引入约束来高效解决该问题。

这项工作的主要贡献在于：(1) 我们首次引入了针对多线性分布数据的全自动分组对应优化方法；(2) 我们证明，与现有的基于PCA的优化方法相比，我们的方法在计算效率上显著更高，并能获得更高质量的对应关系。

2. 相关工作

基于模板的人脸对应计算：我们的方法与旨在计算形状集之间对应关系的方法相关。虽然已有许多方法可用于建立任意类别形状的对应关系，但我们专注于三维人脸配准方法。给定一组稀疏的三维地标点，Mpiperis *et al*.[24] 使用弹性可变形人脸模型配准不同表情下的三维人脸。Passalis *et al*.[26] 通过求解二阶微分方程，将带注释的人脸模型拟合到扫描数据上。Huang *et al*.[17] 将人脸划分为多个部分，并对每个部分进行变形以拟合输入人脸。Guo *et al*.[14] 使用薄板样条引导的模板拟合来配准三维人脸扫描。Pan *et al*.[25] 使用稀疏可变形模型进行配准。他们在一组已配准的人脸上学习字典，并通过将对应关系限制为所学字典的稀疏表示来配准新的人脸。Salazar *et al*.[27] 使用混合形状模型拟合表情，随后使用非刚性迭代最近点方法进行模板拟合，以获取面部细节。

所有这些方法都能找到良好的对应关系，但其中没有一种旨在生成对统计建模最优的配准结果。请注意，这些方法中的任何一种都可以用于初始化我们的优化方法。

统计面部模型：给定一组具有完全对应关系的三维形状，各种方法可以执行统计分析。我们将讨论重点放在三维面部的多线性形状空间上。Vlasic *et al*等人[30]使用多线性模型处理人类面部数据库，该模型将面部形状、表情和视位解耦，以在二维视频之间迁移面部表现。Mpiperis *et al*等人[24]使用多线性模型进行身份和表情识别。

Yang *et al*. [31] 从2D视频重建3D人脸形状，并利用身份与表情变化的解耦来修改视频中的身份或表情。Bolkart和Wuhrer [2] 使用多线性模型对大量运动中的3D人脸数据库进行配准，并对配准结果进行分析。Brunton *et al*. [5] 学习多个局部多线性模型，并利用这些模型从含噪声且有遮挡的人脸扫描中重建模型。

由于这些方法对多个身份和表情的三维人脸使用多线性模型，因此它们采用了与我们的方法相同的模型。然而，它们均未旨在利用所学模型优化对应关系。

配准优化：虽然机器学习中的一些先前工作探索了联合学习模型与对应信息的想法（*e.g*。[4, 29, 18]），但我们的方法与那些旨在联合优化一组3D形状的配准和学习到的统计模型的方法最为相关。Kotcheff和Taylor [21]提出了一种基于PCA模型的群体wise对应优化，该模型明确偏好紧凑模型。Davies *et al*。[10，第4章]概述了不同的目标函数用于对应优化，并激发了一种信息论目标函数，该函数最小化数据的描述长度。最小描述长度方法的基本概念是最小化从发送者传输到接收者的消息长度。他们使用PCA模型对数据进行编码，并改变对应关系，使得描述模型和编码数据所需的位数最小。Davies *et al*。[9]表明，MDL在医学数据集上优于最先进的配准方法。Gollmer *et al*。[13]比较了不同的目标函数。他们表明，虽然协方差矩阵的行列式更容易优化，但结果与MDL产生的结果相当。

所有这些方法都使用单一的线性PCA模型对数据进行建模。相比之下，Burghard *et al*. [6] 采用了基于部件的线性模型，而Chen *et al*. [8] 则使用非线性核PCA对数据进行建模。Hirshberg *et al*. [16] 提出了一种专门针对人体形状的基于骨架的方法，以联合优化配准和统计模型。

这些方法均无法建模具有不同身份和表情的3D人脸。为此，我们提出了首个面向多线性分布数据的组对应优化方法。此外，多数方法假设对象为封闭流形，而我们的方法可处理具有多个边界的流形。

3. 多线性形状空间

本节介绍了多线性模型以及用于推导该模型的不同张量分解方法。多线性模型能够有效地对面部由于身份和表情引起的统计变化进行建模，因为它将这两种形状变化解耦开来。

3.1. Multilinear model

Given a set of registered and spatially aligned 3D face scans of d_2 identities in d_3 expressions each, every face is represented by a vector $\mathbf{f} = (x_1, y_1, z_1, \cdots, x_n, y_n, z_n)^T$ that consists of n vertices (x_i, y_i, z_i). We center each face by subtracting the mean over all training faces $\bar{\mathbf{f}}$ and arrange the centered faces in a 3-mode tensor $\mathcal{A} \in \mathbb{R}^{3n \times d_2 \times d_3}$, where modes describe the different axes of a tensor. The data are placed within $\mathcal{A}$, such that the vertex coordinates are associated with the first mode, the different identities with the second mode, and the different expressions with the third mode of $\mathcal{A}$. The decomposition of $\mathcal{A}$ into

$$\mathcal{A} = \mathcal{M} \times_2 \mathbf{U}_2 \times_3 \mathbf{U}_3, \tag{1}$$

where $\times_n$ denotes the n-th mode product, results in a tensor $\mathcal{M} \in \mathbb{R}^{3n \times m_2 \times m_3}$ called multilinear model, and orthogonal factor matrices $\mathbf{U}_2 \in \mathbb{R}^{d_2 \times m_2}$ and $\mathbf{U}_3 \in \mathbb{R}^{d_3 \times m_3}$. The n-th mode product $\mathcal{M} \times_n \mathbf{U}_n$ of tensor $\mathcal{M}$ with matrix $\mathbf{U}_n$ replaces each vector $\mathbf{m} \in \mathbb{R}^{m_n}$ aligned with i-th mode by $\mathbf{U}_n\mathbf{m} \in \mathbb{R}^{d_n}$. The multilinear model represents a registered 3D face $\mathbf{f} \in \mathbb{R}^{3n}$ as

$$\mathbf{f} \approx \bar{\mathbf{f}} + \mathcal{M} \times_2 \mathbf{w}_2^T \times_3 \mathbf{w}_3^T, \tag{2}$$

where $\mathbf{w}_2 \in \mathbb{R}^{m_2}$ and $\mathbf{w}_3 \in \mathbb{R}^{m_3}$ are the identity and expression coefficients.

3.2. Tensor decompositions

The decomposition of $\mathcal{A}$ in Equation 1 is called Tucker decomposition. The goal is to find the best Tucker decomposition with a lower-dimensional tensor $\mathcal{M}$ that is as close as possible to $\mathcal{A}$. The quality of the tensor approximation is measured by the norm of the residual. Computing the best Tucker decomposition is NP-hard [15]. Furthermore, in contrast to decomposing a matrix into orthogonal matrices (computed using singular value decomposition (SVD)), Tucker decompositions are not unique. An exact Tucker decomposition can be computed if $m_n = rank(\mathbf{A}_{(n)})$ for all n. Here, $\mathbf{A}_{(n)}$ denotes the matrix unfolding of $\mathcal{A}$ in the direction of n-th mode (all vectors in the direction of the n-th mode form the columns of $\mathbf{A}_{(n)}$). If $m_n < rank(\mathbf{A}_{(n)})$ for at least one n, the decomposition approximates $\mathcal{A}$.

The following describes different methods to compute the Tucker decomposition. Section 5.1 evaluates for each method its ability to reconstruct unseen data when applied for model fitting. We compare the three tensor decompositions described by Kolda and Bader [20], namely: higher order SVD (HOSVD) [22], higher order orthogonal iteration (HOOI) [22], and a Newton-Grassmann optimization approach [12]. All these methods compute a Tucker decomposition for given maximum mode ranks m_2 and m_3.

HOSVD: HOSVD is a higher-order generalization of matrix SVD. To compute the matrices $\mathbf{U}_n$, a matrix SVD is performed as $\mathbf{A}_{(n)} = \mathbf{U}_n\mathbf{S}_n\mathbf{V}_n^T$, where $\mathbf{U}_n \in \mathbb{R}^{d_n \times d_n}$ contains the left singular vectors of $\mathbf{A}_{(n)}$. Truncating columns then reduces the dimensions of identity and expression space. The multilinear model is then computed as $\mathcal{M} = \mathcal{A} \times_2 \mathbf{U}_2^T \times_3 \mathbf{U}_3^T$. Even for given m_2 and m_3, the truncated HOSVD does not give an optimal approximation of $\mathcal{A}$.

HOOI: Initialized by HOSVD, this method iteratively optimizes the Tucker decomposition. Within each iteration, both factor matrices are updated by fixing one and updating the other. That is, for a fixed mode-2 factor matrix, a tensor $\mathcal{X} = \mathcal{A} \times_2 \mathbf{U}_2^T$ is computed, and $\mathbf{U}_3$ is updated by the m_3 left singular vectors of $\mathbf{X}_{(3)}$. A similar computation is performed for a fixed mode-3 factor matrix. While HOOI gives a better approximation of $\mathcal{A}$ than HOSVD, it does not necessarily find a stationary point.

Newton-Grassmann optimization: Initialized by HOSVD, the Newton-Grassmann optimization approach constrains each factor matrix to a Grassmannian manifold, an equivalence class of orthogonal matrices. The Tucker decomposition is then computed by a non-linear Newton method on the product of two Grassmannian manifolds. This method converges to a stationary point.

The evaluation of the different tensor decompositions shows that applied to reconstructing unseen face data, they perform almost identical (see Section 5.1). Since HOSVD is the most efficient approach, in the following, we use HOSVD to learn the multilinear model.

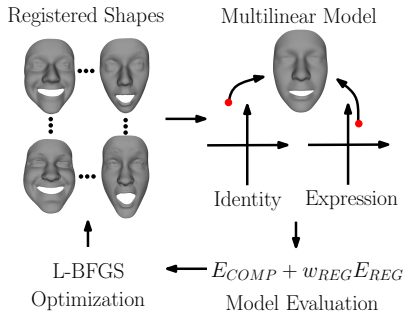


Figure 1. Overview of the iterative multilinear registration.

4. Groupwise correspondence optimization

This section introduces the concept of groupwise correspondence optimizations and describes our approach for multilinearly distributed data. Given a set of shapes in correspondence, groupwise correspondence optimization minimizes an objective function that measures the quality of the correspondence depending on all shapes. Using a statistical model that describes the variation of the shapes, the objective function measures favorable properties of the model.

For PCA models, Kotcheff and Taylor [21] choose the objective function to be the determinant of the covariance matrix, which explicitly favors the induced linear statistical

3.1. 多重线性模型

给定一组已配准且空间对齐的3D人脸扫描，包含d_2个身份和d_3种表情，每张人脸由向量$\mathbf{f} = (x_1, y_1, z_1, \cdots, x_n, y_n, z_n)^T$表示，该向量由$n$个顶点组成（$x_i, y_i, z_i$）。我们通过减去所有训练人脸f的均值来对每张人脸进行中心化处理，并将中心化后的人脸排列成3模张量$\mathcal{A} \in \mathbb{R}^{3n \times d_2 \times d_3}$，其中各模描述张量的不同轴。数据置于$\mathcal{A}$中，使得顶点坐标与第一模关联，不同身份与第二模关联，不同表情与第三模关联（$\mathcal{A}$）。将$\mathcal{A}$分解为

$$\mathcal{A} = \mathcal{M} \times_2 \mathbf{U}_2 \times_3 \mathbf{U}_3, \tag{1}$$

其中 $\times_n$ 表示第 n 个模态乘积，结果得到一个张量 $\mathcal{M} \in \mathbb{R}^{3n \times m_2 \times m_3}$，称为多线性模型，以及正交因子矩阵 $\mathrm{U}_2 \in \mathbb{R}^{d_2 \times m_2}$ 和 $\mathrm{U}_3 \in \mathbb{R}^{d_3 \times m_3}$。张量 $\mathcal{M}$ 与矩阵 U_n 的第 n 个模态乘积 $\mathcal{M} \times_n \mathrm{U}_n$，将每个与第 i 个模态对齐的向量 $\mathrm{m} \in \mathbb{R}^{m_n}$ 替换为 $\mathrm{U}_n\mathrm{m} \in \mathbb{R}^{d_n}$。多线性模型将注册的3D人脸 $\mathrm{f} \in \mathbb{R}^{3n}$ 表示为

$$\mathbf{f} \approx \bar{\mathbf{f}} + \mathcal{M} \times_2 \mathbf{w}_2^T \times_3 \mathbf{w}_3^T, \tag{2}$$

其中$\mathrm{w}_2 \in \mathbb{R}^{m_2}$和$\mathrm{w}_3 \in \mathbb{R}^{m_3}$为身份系数和表情系数。

3.2. 张量分解

公式1中$\mathcal{A}$的分解称为Tucker分解。目标是找到具有更低维张量$\mathcal{M}$的最佳Tucker分解，使其尽可能接近$\mathcal{A}$。张量近似的质量通过残差的范数来衡量。计算最佳Tucker分解是NP难题[15]。此外，与将矩阵分解为正交矩阵（使用奇异值分解（SVD）计算）不同，Tucker分解不是唯一的。如果对于所有n，$m_n = rank(\mathrm{A}_{(n)})$成立，则可以计算精确的Tucker分解。这里，$\mathrm{A}_{(n)}$表示$\mathcal{A}$沿$n$阶模方向的矩阵展开（沿$n$阶模方向的所有向量构成$\mathrm{A}_{(n)}$的列）。如果至少存在一个$n$使得$m_n < rank(\mathrm{A}_{(n)})$成立，则该分解近似于$\mathcal{A}$。

下面描述了计算Tucker分解的不同方法。第5.1节评估了每种方法在模型拟合时重建未见数据的能力。我们比较了Kolda和Bader [20]描述的三种张量分解方法，即：高阶SVD（HOSVD）[22]、高阶正交迭代（HOOI）[22]以及Newton-Grassmann优化方法 [12]。所有这些方法在给定最大模秩 m_2 和 m_3 的情况下计算Tucker分解。

HOSVD：HOSVD是矩阵SVD的高阶推广。为了计算矩阵U_n，需要执行矩阵SVD。

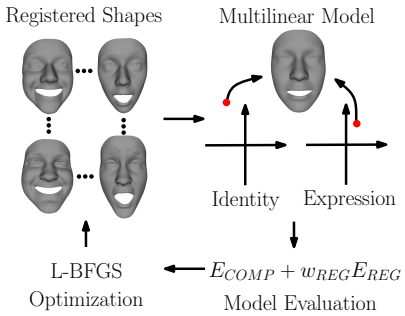


图1. 迭代多重线性配准的概览。

执行得到 $\mathrm{A}_{(n)} = \mathrm{U}_n\mathrm{S}_n\mathrm{V}_n^T$，其中 $\mathrm{U}_n \in \mathbb{R}^{d_n \times d_n}$ 包含 $\mathrm{A}_{(n)}$ 的左奇异向量。截断列后，身份和表情空间的维度随之降低。多线性模型随后计算为 $\mathcal{M} = \mathcal{A} \times_2 \mathrm{U}_2^T \times_3 \mathrm{U}_3^T$。即使给定 m_2 和 m_3，截断的HOSVD也不能给出 $\mathcal{A}$ 的最优近似。

HOOI：由HOSVD初始化，该方法迭代优化Tucker分解。在每次迭代中，通过固定一个因子矩阵并更新另一个来更新两个因子矩阵。即，对于固定的模式2因子矩阵，计算张量$\mathcal{X} = \mathcal{A} \times_2 \mathrm{U}_2^T$，并通过$\mathrm{X}_{(3)}$的$m_3$个左奇异向量更新$\mathrm{U}_3$。对于固定的模式3因子矩阵，执行类似的计算。虽然HOOI比HOSVD能更好地逼近$\mathcal{A}$，但它不一定能找到驻点。

牛顿-格拉斯曼优化：由HOSVD初始化，牛顿-格拉斯曼优化方法将每个因子矩阵约束在格拉斯曼流形上，即正交矩阵的等价类。随后，通过在两个格拉斯曼流形的乘积上运用非线性牛顿法计算Tucker分解。该方法收敛至一个驻点。

对不同张量分解的评估表明，在重建未见人脸数据时，它们的性能几乎相同（见第5.1节）。由于HOSVD是最有效的方法，因此在后续工作中，我们使用HOSVD来学习多线性模型。

4. 分组对应优化

本节介绍分组对应优化的概念，并描述我们针对多线性分布数据的方法。给定一组处于对应关系的形状，分组对应优化最小化一个目标函数，该函数根据所有形状度量对应的质量。利用描述形状变异的统计模型，该目标函数度量模型的有利属性。

对于PCA模型，Kotcheff和Taylor [21]选择目标函数为协方差矩阵的行列式，这明确有利于诱导的线性统计

model to be compact. The compactness of a linear statistical model can be maximized by minimizing the variability of the model, measured by the trace of the covariance matrix.

Compactness measures the variability captured by a model. A compact model can describe instances of a given dataset with the minimum number of parameters and has minimal variance. For models of different compactness that describe the same data, the model with higher compactness and hence lower variance is favorable. It has been shown that minimizing the variance of a PCA model performs similarly to information theoretic approaches that aim at minimizing the description length of the model [13].

Inspired by these previous works, we develop the first MDL-based optimization approach for multilinear models. This extension is challenging because the notion of compactness needs to be extended to multilinear models, where optimal tensor approximation is NP-hard. For 3D face data, a further challenge arises from manifold boundaries. Figure 1 gives an overview of our multilinear optimization approach. Given a set of 3D faces of different identities performing different expressions with an initial correspondence, we iteratively optimize the correspondence. We compute a multilinear model on the registered data, and iteratively improve the model. In each iteration, the quality of the model is measured using a groupwise objective function (Section 4.1). The registered shapes are represented using a continuous parametrization (Section 4.2), and the objective function is optimized in parameter space with a quasi-Newton method (Section 4.3).

4.1. Multilinear objective function

Our groupwise objective function consists of two parts: a compactness energy E_{COMP}, and a regularization energy E_{REG}. We therefore aim to minimize

$$E = E_{COMP} + w_{REG} E_{REG}, \tag{3}$$

where w_{REG} is a weight that controls the influence of the regularization. We now describe both terms in more detail.

Compactness: The compactness of a multilinear model can be measured as the percentage of data variability captured in the first k components of each mode, where $k = 1, \ldots, \max(d_2, d_3)$ [2]. Compactness is maximized by a sparse model that captures all of the variability in few components. To encourage a sparse model, we introduce an energy on the variability of the identity and expression subspaces. Like Kotcheff and Taylor [21], we choose a log-sum penalty function, as log-sum functions are known to encourage sparsity by heavily punishing small values [7]. That is, we aim to minimize

$$E_{COMP} = \frac{1}{d_2}\sum_{i=1}^{d_2} \ln(\lambda_i^{(2)} + \delta_2) + \frac{1}{d_3}\sum_{i=1}^{d_3} \ln(\lambda_i^{(3)} + \delta_3), \tag{4}$$

where $\lambda_i^{(n)}$ denotes the i-th eigenvalue of the mode-n covariance matrix. Small regularization constants δ_n are used to avoid singularities of E_{COMP} for vanishing eigenvalues. Equivalent to HOSVD, the mode-2 and mode-3 covariance matrices are computed as $\frac{1}{d_3}\mathbf{A}_{(2)}\mathbf{A}_{(2)}^T$ and $\frac{1}{d_2}\mathbf{A}_{(3)}\mathbf{A}_{(3)}^T$.

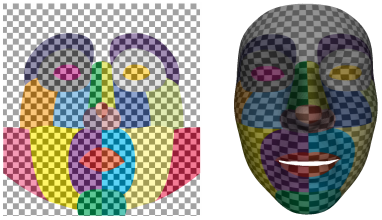


Figure 2. Initial surface parametrization of the 3D face template. Left: 2D parameter domain. Right: 3D parametrization.

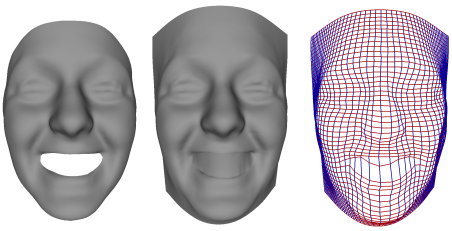


Figure 3. Parametrization for one shape. Left: initialization. Middle: thin-plate spline. Right: (u, v)-parameter lines.

The energy E_{COMP} is minimized by moving points within the continuous surface of each shape. Since the computation of the covariance only considers a discrete number of points instead of the continuous surface, E_{COMP} can be minimized by moving points away from complex geometric regions with high variability.

Regularization: To avoid undersampling in these regions, Davies *et al.* [10] approximate the integral of the continuous covariance matrix by weighting the points by their surrounding surface area. Since this does not always prevent the undersampling [13], as done in Burghard *et al.* [6], we use a regularization within the objective function. The regularization term for each shape is a bi-Laplacian of the form

$$E_{REG} = \frac{1}{n}\sum_{k=1}^{n} \left\| U^2(\mathbf{v}_k(\mathbf{x})) \right\|^2, \tag{5}$$

where $\mathbf{v}_k(\mathbf{x})$ denotes the k-th vertex of shape $\mathbf{x}$. The double-umbrella operator $U^2(\mathbf{p})$ is the discrete bi-Laplacian approximation [19] computed by

$$U^2(\mathbf{p}) = \frac{1}{|N(\mathbf{p})|}\sum_{\mathbf{p}_r \in N(\mathbf{p})} U(\mathbf{p}_r) - U(\mathbf{p}), \tag{6}$$

where $N(\mathbf{p})$ denotes the set of neighbors of vertex $\mathbf{p}$ within the mesh, and $U(\mathbf{p}) = 1/|N(\mathbf{p})| \sum_{\mathbf{p}_r \in N(\mathbf{p})} \mathbf{p}_r - \mathbf{p}$. The bi-Laplacian regularizer encourages the points to be regularly distributed over the mesh and prevents fold-overs.

使模型保持紧凑。线性统计模型的紧凑性可通过最小化模型的变异性来最大化，变异性以协方差矩阵的迹来衡量。

紧凑性衡量模型所捕获的变异性。一个紧凑的模型能够以最少的参数数量描述给定数据集中的实例，并且具有最小的方差。对于描述相同数据但紧凑性不同的模型，紧凑性更高因而方差更低的模型更为可取。已有研究表明，最小化PCA模型的方差与旨在最小化模型描述长度的信息论方法表现相似[13]。

受上述工作启发，我们首次提出了针对多线性模型的基于MDL的优化方法。这一扩展颇具挑战性，因为紧致性的概念需要扩展到多线性模型，而其中最优张量逼近是NP难题。对于三维人脸数据，流形边界带来了又一挑战。图1展示了我们多线性优化方法的概览。给定一组具有不同身份、执行不同表情且具有初始对应关系的三维人脸，我们迭代优化对应关系。我们在配准后的数据上计算多线性模型，并迭代改进该模型。在每次迭代中，使用组间目标函数（第4.1节）衡量模型质量。配准后的形状采用连续参数化表示（第4.2节），目标函数在参数空间中通过拟牛顿方法进行优化（第4.3节）。

4.1. 多重线性目标函数

我们的分组目标函数由两部分组成：紧致能量 E_{COMP} 和正则化能量 E_{REG}。因此，我们旨在最小化

$$E = E_{COMP} + w_{REG} E_{REG}, \tag{3}$$

其中 w_{REG} 是控制正则化影响程度的权重。下面我们将详细描述这两个项。

紧凑性：多线性模型的紧凑性可以通过每个模态的前 k 个分量所捕获的数据变异性百分比来衡量，其中 $k = 1, \ldots, \max(d_2, d_3)$ [2]。稀疏模型通过在少量分量中捕获全部变异性来实现紧凑性的最大化。为了鼓励模型稀疏，我们在身份和表情子空间的变异性上引入了一种能量项。与Kotcheff和Taylor [21]类似，我们选择对数求和惩罚函数，因为已知对数求和函数通过严厉惩罚小值来促进稀疏性[7]。也就是说，我们的目标是最小化。

$$E_{COMP} = \frac{1}{d_2}\sum_{i=1}^{d_2} \ln(\lambda_i^{(2)} + \delta_2) + \frac{1}{d_3}\sum_{i=1}^{d_3} \ln(\lambda_i^{(3)} + \delta_3), \tag{4}$$

其中，$\lambda_i^{(n)}$ 表示模态-n 协方差矩阵的第 i 个特征值。使用小的正则化常数 δ_n

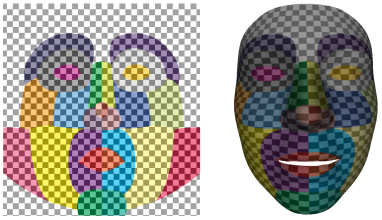


图2. 3D人脸模板的初始曲面参数化。左：2D参数域。右：3D参数化。

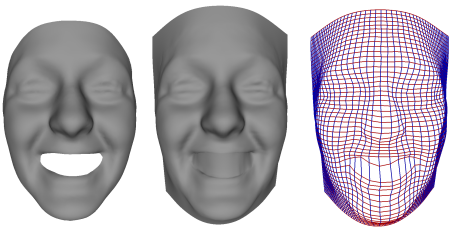


图3. 单个形状的参数化。左：初始化。中：薄板样条。右：(u, v)参数线。

为了避免特征值趋零时E_{COMP}出现奇异，计算与HOSVD等效的模态2和模态3协方差矩阵：$\frac{1}{d_3}\mathrm{A}_{(2)}\mathrm{A}_{(2)}^T$ 和$\frac{1}{d_2}\mathrm{A}_{(3)}\mathrm{A}_{(3)}^T$。

能量 E_{COMP} 通过在每个形状的连续表面内移动点来实现最小化。由于协方差的计算仅考虑离散数量的点而非连续表面，E_{COMP} 可以通过将点移离具有高可变性的复杂几何区域来实现最小化。

正则化：为避免这些区域中的欠采样，Davies *et al*。[10] 通过按点周围表面积加权来近似连续协方差矩阵的积分。由于这并不总能防止欠采样[13]，如Burghard *et al*。[6]所做，我们在目标函数中使用了正则化。每个形状的正则化项是如下形式的双拉普拉斯算子

$$E_{REG} = \frac{1}{n}\sum_{k=1}^{n} \left\| U^2(\mathbf{v}_k(\mathbf{x})) \right\|^2, \tag{5}$$

其中 $\mathrm{v}_k(\mathrm{x})$ 表示形状 x 的第 k 个顶点。双重伞算子 $U^2(\mathrm{p})$ 是由下式计算得到的离散双拉普拉斯近似[19]：

$$U^2(\mathbf{p}) = \frac{1}{|N(\mathbf{p})|}\sum_{\mathbf{p}_r \in N(\mathbf{p})} U(\mathbf{p}_r) - U(\mathbf{p}), \tag{6}$$

其中$N(\mathrm{p})$表示网格中顶点p的邻居集合，且$U(\mathrm{p}) = 1/|N(\mathrm{p})| \sum_{\mathbf{p}_r \in N(\mathbf{p})} \mathrm{p}_r - \mathrm{p}$。双拉普拉斯正则化器促使点均匀分布在网格上，并防止折叠。

4.2. Parametrization

The registration is optimized by moving points in the surface of each face. Since the surface of the face is 2-dimensional, moving points within the surface can be done by re-parametrization. This requires an initial parametrization together with a continuous mapping from parameter space to the surface of each face. We compute an initial registration for a database of 3D faces using template fitting, and additionally unwrap the 3D template mesh in 2D parameter space to compute an initial discrete parametrization with parameters $\mathbf{t}_i \in \mathbb{R}^2$. The embedding in 2D is chosen to minimize distortions of angles and areas. Each parameter $\mathbf{t}_i$ is mapped to the mesh vertex $\mathbf{v}_i = (x_i, y_i, z_i) \in \mathbb{R}^3$. Figure 2 visualizes the initial parametrization in 2D parameter space (left) and mapped on the 3D surface (right). Due to the full correspondence of all face shapes, this discrete parametrization is the same for all shapes of the database.

With this discrete embedding in parameter space, a continuous mapping Φ is computed that maps parameters $\boldsymbol{\alpha} = (u, v) \in \mathbb{R}^2$ into the surface of the shape. A thin-plate spline [11] defines this mapping, computed as

$$\Phi(\boldsymbol{\alpha}) = \mathbf{c} + \mathbf{A}\boldsymbol{\alpha} + \mathbf{W}^T(\sigma(\boldsymbol{\alpha} - \mathbf{t}_1), \ldots, \sigma(\boldsymbol{\alpha} - \mathbf{t}_n))^T, \quad (7)$$

where $\mathbf{c} \in \mathbb{R}^3$, $\mathbf{A} \in \mathbb{R}^{3\times 2}$, and $\mathbf{W} \in \mathbb{R}^{n\times 3}$ are the parameters of the mapping, and where $\sigma : \mathbb{R}^2 \to \mathbb{R}$ is the function

$$\sigma(\mathbf{h}) = \begin{cases} \|\mathbf{h}\|^2 \log(\|\mathbf{h}\|) & \|\mathbf{h}\| > 0, \\ 0 & \|\mathbf{h}\| = 0. \end{cases} \quad (8)$$

The surface of Φ interpolates all vertices of the shape ($\Phi(\mathbf{t}_i) = \mathbf{v}_i$) and gives the surface with the minimum bending energy. Figure 3 shows one initially registered shape (left) together with the computed continuous thin-plate spline visualized as densely approximated mesh (middle) and (u, v)-parameter lines (right). The evaluation of Φ at parameters $\boldsymbol{\alpha}$, where u (respectively v) is fixed and v (respectively u) is varied by a fixed discrete step size, gives one (u, v)-parameter line. While the spline interpolates the geometry of the initial shape, it gives a reasonable extrapolation of the shape beyond the outer border of the face.

4.3. Optimization

The objective function E in Equation 3 is non-linear. Due to the choice of the parametrization, E is analytically differentiable with respect to $\boldsymbol{\alpha}$. The supplementary material gives the full analytical gradient. We minimize E using L-BFGS [23], a quasi-Newton method with linear constraints. These linear constraints allow for each vertex in parameter space to specify a valid rectangular area.

Boundary constraints: For meshes with boundary, E_{COMP} is minimized if the entire surface collapses into a single point. Hence, boundary conditions need to be enforced. Face shapes have two boundaries, an inner boundary at the mouth and an outer boundary at the end of the acquired scan. Since landmarks are used during the initial registration, the inner boundary at the mouth is registered well. To avoid points that move from the lower to the upper lip or vice versa, we fix the points in the 1-ring neighborhood of the mouth boundary during optimization. Since the outer boundary is not registered well as scans in the database are cropped inconsistently, we allow limited movement for points in the 1-ring neighborhood of the outer boundary. Specifically, the movement is restricted to at most 20 mm.

Optimization schedule: Optimizing for the parameters of all shapes at the same time is not feasible for a large population of shapes due to the large number of parameters ($d_2 d_3 2n$). Instead, we only optimize the parameters of each shape individually as proposed by Davies *et al.* [10, Chapter 7.1.1]. This optimization is performed for all shapes of the database during each iteration. Note that E still depends on all shapes for this shape-wise optimization, and the method therefore still optimizes the groupwise correspondence. To avoid bias towards any shape, the order of the shapes is randomly permuted for each iteration step. Since the rigid alignment of the shapes depends on the correspondence, during optimization of one shape, the alignment is updated after a few optimization steps.

Computational complexity: The computational complexity of one optimization step is $O(nd_2^2 d_3 + nd_2 d_3^2)$ (see supplementary material for details). As shown in the following section, our approach is significantly more efficient than existing PCA-based MDL approaches.

5. Evaluation

This section evaluates three different tensor decompositions and our model optimization approach.

Data: For evaluation, we use models of the BU-3DFE [32] and Bosphorus [28] databases. BU-3DFE contains 3D face scans in neutral expression and in six prototypic expressions. Bosphorus covers the six prototypic expressions and a subset of up to 28 action units per subject. Since both databases are acquired with different scanner systems, the resulting scans have different resolution and noise characteristics. We register the face scans with a template fitting method [27] using the provided landmarks.

For BU-3DFE we use 50 randomly chosen identities in 7 expressions: neutral and the highest level of each expression. For Bosphorus we use all 65 identities that are present in all 7 expressions. In the following, we call these subsets BU-3DFE subset and Bosphorus subset, respectively.

Model quality: We quantitatively evaluate the quality of the optimization with the widely used measures compactness, generalization and specificity [10, Chapter 9.2]. The

4.2. 参数化

配准通过移动每张人脸表面上的点来优化。由于人脸表面是二维的，在表面内移动点可以通过重新参数化来实现。这需要初始参数化以及从参数空间到每张人脸表面的连续映射。我们使用模板拟合为三维人脸数据库计算初始配准，并在二维参数空间中对三维模板网格进行展开，以计算初始离散参数化，其参数为$\mathbf{t}_i \in \mathbb{R}^2$。二维嵌入的选择旨在最小化角度和面积的畸变。每个参数$\mathbf{t}_i$被映射到网格顶点$\mathbf{v}_i = (x_i, y_i, z_i) \in \mathbb{R}^3$。图2可视化了二维参数空间中的初始参数化（左）及其映射到三维表面上的结果（右）。由于所有人脸形状完全对应，这种离散参数化对数据库中的所有形状都是相同的。

借助参数空间中的这种离散嵌入，计算出一个连续映射Φ，该映射将参数$\boldsymbol{\alpha} = (u, v) \in \mathbb{R}^2$映射到形状表面。薄板样条[11]定义了该映射，计算如下：

$$\Phi(\boldsymbol{\alpha}) = \mathbf{c} + \mathbf{A}\boldsymbol{\alpha} + \mathbf{W}^T(\sigma(\boldsymbol{\alpha} - \mathbf{t}_1), \ldots, \sigma(\boldsymbol{\alpha} - \mathbf{t}_n))^T, \quad (7)$$

其中$\mathbf{c} \in \mathbb{R}^3$、$\mathbf{A} \in \mathbb{R}^{3\times 2}$和$\mathbf{W} \in \mathbb{R}^{n\times 3}$是映射的参数，且$\sigma\colon \mathbb{R}^2 \to \mathbb{R}$是函数

$$\sigma(\mathbf{h}) = \begin{cases} \|\mathbf{h}\|^2 \log(\|\mathbf{h}\|) & \|\mathbf{h}\| > 0, \\ 0 & \|\mathbf{h}\| = 0. \end{cases} \quad (8)$$

Φ的表面插值了形状的全部顶点（$\Phi(\mathbf{t}_i) = \mathbf{v}_i$），并给出了具有最小弯曲能量的表面。图3展示了一个初始配准的形状（左），以及计算得到的连续薄板样条，该样条以密集逼近网格（中）和(u, v)参数线（右）的形式可视化。在参数$\boldsymbol{\alpha}$处对Φ的评估，其中u (和v)分别固定，而v和(分别以固定的离散步长变化，得到一条(u, v)参数线。虽然样条插值了初始形状的几何形状，但它对超出面部外边界的形状给出了合理的 extrapolation（外推）。

4.3. 优化

方程3中的目标函数E是非线性的。由于参数化选择的缘故，E关于$\boldsymbol{\alpha}$具有解析可微性。补充材料给出了完整的解析梯度。我们使用L-BFGS[23]（一种带线性约束的拟牛顿法）来最小化E。这些线性约束允许参数空间中的每个顶点指定一个有效的矩形区域。

边界约束：对于带边界的网格，若整个表面坍缩为一个单点。因此，需要强制施加边界条件。面部形状有两个边界：一个内边界位于嘴部，一个外边界位于采集扫描的末端。由于在初始配准过程中使用了地标，嘴部的内边界配准效果良好。为避免点从下唇移动到上唇或反之，我们在优化过程中固定嘴部边界1环邻域内的点。由于数据库中的扫描裁剪不一致，外边界配准效果不佳，我们允许外边界1环邻域内的点有有限的移动。具体而言，移动限制在最多20毫米。

优化调度：由于参数数量庞大（$d_2 d_3 2n$），同时优化所有形状的参数对于大规模形状群体而言并不可行。因此，我们仅按照Davies的建议逐一优化每个形状的参数*et al*。[10，第7.1.1节]。在每次迭代中，对数据库中的所有形状执行该优化。请注意，E在此逐形状优化中仍依赖于所有形状，因此该方法仍能优化群体对应关系。为避免对任何形状产生偏差，每次迭代时随机排列形状的顺序。由于形状的刚性对齐依赖于对应关系，在对一个形状进行优化期间，对齐会在若干优化步骤后更新。

计算复杂度：一个优化步骤的计算复杂度为$O(nd_2^2 d_3 + nd_2 d_3^2)$ (详见补充材料)。如下一节所示，我们的方法比现有的基于PCA的MDL方法效率显著更高。

5. 评估

本节评估三种不同的张量分解方法以及我们的模型优化方法。

数据：为了评估，我们使用了BU-3DFE [32]和Bosphorus [28]数据库中的模型。BU-3DFE包含中性表情和六种典型表情的3D人脸扫描。Bosphorus涵盖六种典型表情以及每个受试者最多28个动作单元的子集。由于两个数据库使用不同的扫描仪系统获取，生成的扫描具有不同的分辨率和噪声特征。我们使用模板拟合方法[27]及提供的标志点对人脸扫描进行配准。

对于BU-3DFE，我们使用7种表情中随机选取的50个身份：中性表情以及每种表情的最高级别。对于Bosphorus，我们使用全部65个在所有7种表情中都出现的身份。在下文中，我们分别将这些子集称为BU-3DFE子集和Bosphorus子集。

模型质量：我们使用广泛采用的紧凑性、泛化性和特异性度量[10, 第9.2章]定量评估优化的质量。

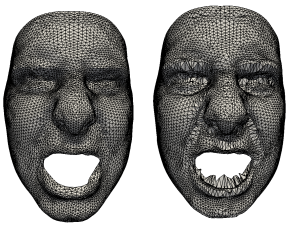


Figure 4. Artifacts obtained by optimizing E_{COMP} without regularization ($w_{REG} = 0$). Left: initial registration. Right: result.

identity and expression spaces should ideally be compact, general and specific.

Generalization measures the ability of the statistical model to represent shapes that are not part of the training. The generalization error is measured in a leave-one-out fashion. For the identity mode, each subject is once fully excluded from training and the resulting model is used to reconstruct all excluded scans. The error is then measured as the average vertex distance between all corresponding vertices. The error for the expression mode is computed accordingly by excluding once each expression.

Specificity measures the ability of the statistical model to only represent valid shapes of the object class. To measure the specificity of the model before and after optimization, we randomly choose 10000 samples in identity and expression space and measure the average vertex distance of the reconstruction to the training data.

Reproducibility: To facilitate evaluating the model for different applications, we make our optimization code and the optimized statistical model available [3].

5.1. Tensor decompositions

We evaluate the different tensor decomposition methods described in Section 3.2 by fitting the resulting multilinear models to unseen 3D face scans. For this, we use a 10-fold cross validation on the registered BU-3DFE scans. We split the database randomly into ten groups, each with the same ratio of male and female subjects, where all scans of one identity belong to the same group. The error is measured as the distance between a vertex in the fitting result and its closest point in the face scan. The error distribution of all three methods is nearly identical. The median vertex error is for HOSVD 1.145 mm, for HOOI 1.144 mm and for the Newton Grassmann method 1.144 mm. Since all methods perform almost the same, we compute the decomposition with HOSVD in the following.

5.2. Influence of regularization

This section evaluates the influence of the regularization E_{REG} on the BU-3DFE subset. The optimization is performed twice, once only optimizing E_{COMP} without E_{REG} and once only optimizing E_{REG} without E_{COMP}. As discussed in Section 4.1, the regularizer is needed to avoid undersampling in regions with high variability and fold-overs. Figure 4 shows the result for one face after only five iterations of optimizing E_{COMP}. When minimizing only E_{COMP}, the optimization moves points away from the eyebrows and around the nose, resulting in sparsely sampled regions. Furthermore, fold-overs at the mouth cause visual artifacts. Optimizing E_{REG} leads to regularly sampled meshes. However, E_{COMP} increases in this case. Minimizing E is therefore a tradeoff between getting a compact model and a regular mesh structure. In the following, we empirically choose $w_{REG} = 0.5$.

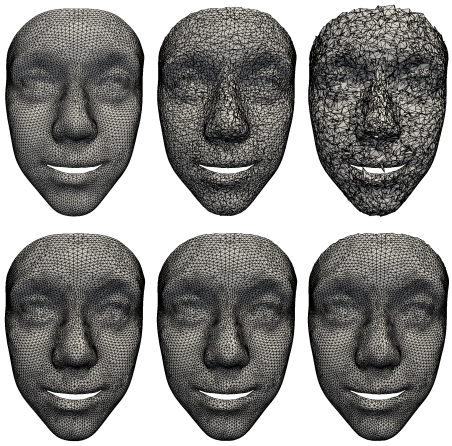


Figure 5. Noise example of the database before (top) and after (bottom) optimization. Left to right: no, low, and high noise.

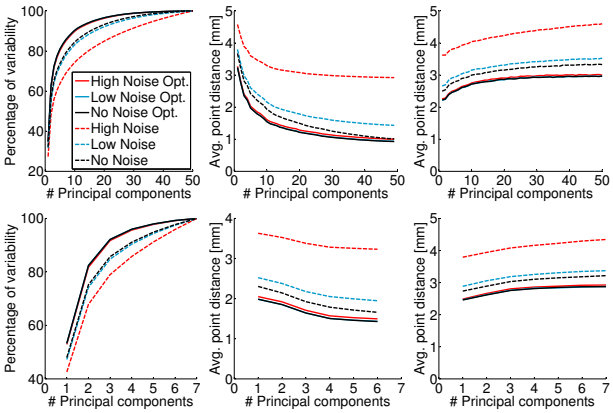


Figure 6. Influence of the initialization for different levels of noise. Left: compactness. Middle: generalization. Right: specificity. Top: identity mode. Bottom: expression mode.

5.3. Influence of initialization

This section evaluates the robustness to noise in the initialization. State-of-the-art registration methods for faces, as used for the initialization of our method, are able to fit the facial surface well with sub-millimeter accuracy, but the result is likely to contain drift within the surface. To simulate noise regarding these methods, we use the initial

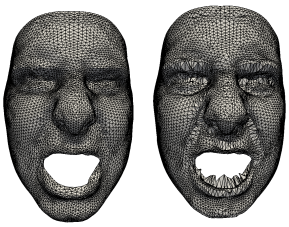


图4：在无正则化（$w_{REG}=0$）条件下优化E_{COMP}所得伪影。左：初始配准。右：结果。

理想情况下，身份与表达空间应当是紧凑、通用且具有针对性的。

泛化衡量的是统计模型表示不属于训练集部分形状的能力。泛化误差采用留一法进行度量。对于身份模式，每个受试者被完全排除在训练之外一次，所得模型用于重建所有被排除的扫描数据。误差随后以所有对应顶点之间的平均顶点距离来衡量。对于表情模式，误差通过每次排除一个表情相应地计算。

特异性衡量统计模型仅表示目标类别有效形状的能力。为在优化前后测量模型的特异性，我们在身份与表情空间中随机选取10000个样本，并测量重建结果到训练数据的平均顶点距离。

可复现性：为便于针对不同应用评估模型，我们公开了优化代码及优化后的统计模型 [3]。

5.1. 张量分解

我们通过将生成的多元线性模型拟合到未见的3D人脸扫描，来评估第3.2节中描述的不同张量分解方法。为此，我们在注册的BU-3DFE扫描上进行了10折交叉验证。我们将数据库随机分为十组，每组中男性和女性受试者的比例相同，且同一身份的所有扫描属于同一组。误差测量为拟合结果中某个顶点与人脸扫描中最近点之间的距离。三种方法的误差分布几乎相同。HOSVD的中位顶点误差为1.145毫米，HOOI为1.144毫米，牛顿格拉斯曼方法为1.144毫米。由于所有方法表现几乎相同，我们在后续使用HOSVD进行分解。

5.2. 正则化的影响

本节评估正则化E_{REG}对BU-3DFE子集的影响。优化执行两次，一次仅优化E_{COMP}而不含E_{REG}，另一次仅优化E_{REG}而不含E_{COMP}。如4.1节所讨论，该正则化项需要用于

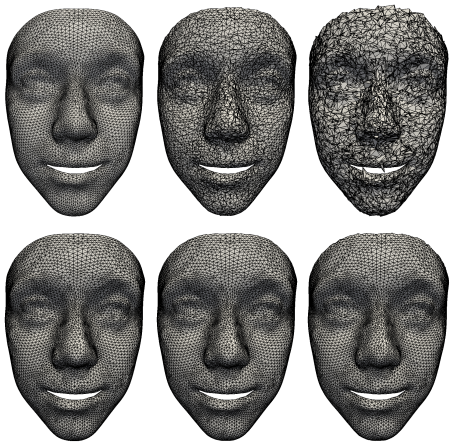


图5. 数据库在优化前（上）和优化后（下）的噪声示例。从左到右：无噪声、低噪声和高噪声。

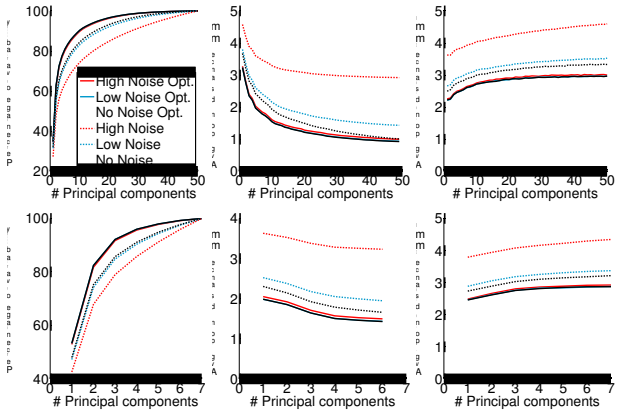


图6. 不同噪声水平下初始化的影响。左：紧致性。中：泛化性。右：特异性。上：身份模式。下：表情模式。

避免在高变异性区域欠采样和折叠。图4展示了仅对E_{COMP}进行五次迭代优化后的一张人脸结果。当仅最小化E_{COMP}时，优化将点从眉毛区域移开并绕鼻子分布，导致稀疏采样区域。此外，嘴部折叠引起视觉伪影。优化E_{REG}可得到规则采样的网格。然而，此时E_{COMP}增加。因此，最小化E是在获得紧凑模型与规则网格结构之间的权衡。接下来，我们经验性地选择$w_{REG}=0.5$。

5.3. 初始化的影响

本节评估初始化过程中对噪声的鲁棒性。用于我们方法初始化的最先进人脸配准方法能够以亚毫米精度很好地拟合面部表面，但结果可能包含表面内的漂移。为模拟这些方法相关的噪声，我们使用初始

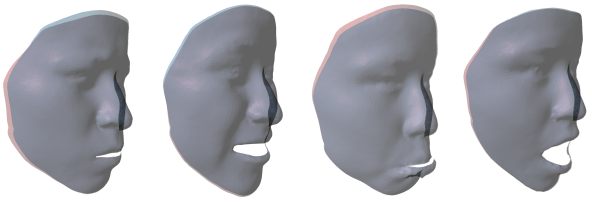


Figure 7. Visual comparison of template fitting [27] (red) and our result (blue) for one subject in four expressions (overlap in gray).

parametrization and add two different levels of noise in the parameter domain. The parameter values of each shape of the BU-3DFE subset are disturbed by random Gaussian noise. Since the 1-ring neighborhood of the mouth boundary is fixed during optimization, these vertices are left without noise. For both noise levels we choose noise with mean zero and standard deviation f times the average 3D edge length. For the lower noise level we choose f to be 0.25, for the higher 0.75, respectively.

The optimization is performed on the BU-3DFE subset, initialized with the noisy registration. The top of Figure 5 shows an example of the database without noise (left), the lower level of noise (middle) and the higher level of noise (right). The average 3D vertex distance of the initial shapes to the noisy shapes over the entire database is 1.11 mm for the lower and 2.50 mm for the higher noise level.

Adding random noise within the surface to each vertex increases the variance in 3D positions and therefore increases the variability of the data. As expected, Figure 6 shows that the compactness of identity mode and expression mode decreases with increasing noise, since the multilinear model captures less variability with the same number of components. Further, the multilinear model becomes less general and less specific. After 15 iterations, the average compactness increases by 3.8% for the low noise level, and by 8.7% for the high noise level, respectively. The average generalization error decreases by $0.58mm$ and $1.65mm$ for the low and high noise level, the average specificity decreases by $0.43mm$ and $1.26mm$ for the low and high noise level. After optimization, the model quality for both levels of noise is comparable to the optimization of the data without noise. Hence, our optimization method effectively reduces variability caused by drift.

5.4. Comparison

This section compares our approach to two state-of-the-art registration methods for 3D faces based on template fitting [27] and PCA-based groupwise correspondence [10].

Template fitting: We compare our optimization to template fitting on the BU-3DFE and Bosphorus subsets. For the two subsets, Figures 8 and 9 show the compactness, generalization and specificity for template fitting and after 15 iterations of the multilinear optimization. For the BU-3DFE subset, the average compactness increases by 3.0%, and the average generalization and specificity decrease by $0.25mm$ and $0.32mm$, respectively. For the Bosphorus subset, the average compactness increases by 1.7%, and the average generalization and specificity decrease by $0.15mm$ and $0.16mm$, respectively.

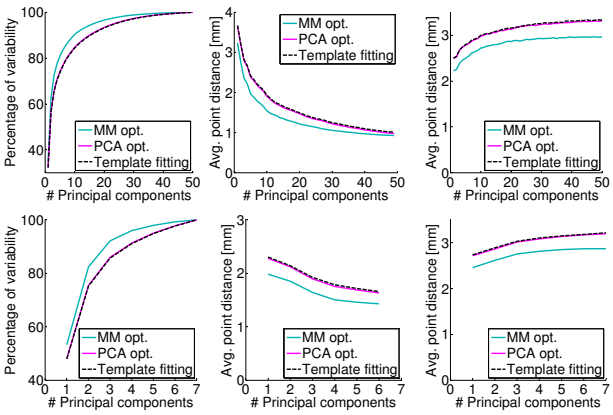


Figure 8. Comparison of template fitting [27], PCA optimization [10] (PCA opt.) and multilinear model optimization (MM opt.) on BU-3DFE subset. Left: compactness. Middle: generalization. Right: specificity. Top: identity mode. Bottom: expression mode.

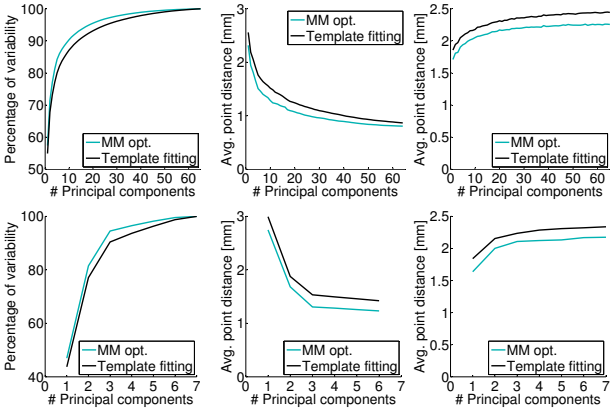


Figure 9. Comparison of template fitting [27] and multilinear model optimization (MM opt.) on Bosphorus subset. Left: compactness. Middle: generalization. Right: specificity. Top: identity mode. Bottom: expression mode.

Figure 7 visually compares the template fitting (red) to our result (blue) for one subject of the BU-3DFE subset. Before optimization, the shape of the outer boundary differs. The optimization decreases the face for the first and fourth expressions at the cheek, for the second expression at the jaw, and for the third expression at the forehead. Expressions one, two and three are extended at the forehead. After 15 iterations, the outer boundaries are similar.

To demonstrate the ability of our method to optimize over large sets of shapes, we consider a second subset of the Bosphorus database consisting of 39 identities performing

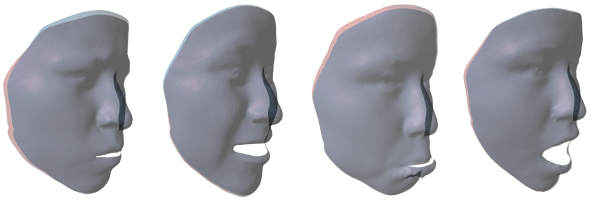


图7. 模板拟合[27]（红色）与我们结果（蓝色）在同一受试者四种表情下的视觉比较（重叠区域为灰色）。

参数化处理，并在参数域中添加两种不同水平的噪声。BU-3DFE子集中每个形状的参数值均受随机高斯噪声干扰。由于优化过程中嘴部边界的1环邻域固定不变，这些顶点不加噪声。对于两种噪声水平，我们选择均值为零、标准差为平均3D边长度的f倍的噪声。较低噪声水平下，f取0.25；较高噪声水平下，.取0.75。

优化在BU-3DFE子集上进行，以带噪声的配准结果初始化。图5顶部展示了数据库中无噪声（左）、较低噪声水平（中）和较高噪声水平（右）的示例。整个数据库上，初始形状与带噪声形状之间的平均三维顶点距离，在较低噪声水平下为1.11毫米，在较高噪声水平下为2.50毫米。

在曲面内对各顶点添加随机噪声会增加3D位置的方差，从而增大数据的变异性。如图6所示，随着噪声增加，身份模式和表情模式的紧凑性下降，因为多线性模型在相同分量数量下捕捉的变异性减少。此外，多线性模型的泛化性和特异性均降低。经过15次迭代后，低噪声水平下的平均紧凑性增加3.8%，高噪声水平下增加8.7%。平均泛化误差分别降低0.58mm和1.65mm，平均特异性分别降低0.43mm和1.26mm。优化后，两种噪声水平下的模型质量与无噪声数据优化结果相当。因此，我们的优化方法有效减少了由漂移引起的变异性。

5.4. 比较

本节将我们的方法与两种基于模板拟合[27]和基于PCA的组对应[10]的最先进三维人脸配准方法进行了比较。

模板拟合：我们在BU-3DFE和Bosphorus子集上将我们的优化方法与模板拟合进行比较。对于这两个子集，图8和图9展示了模板拟合以及多线性优化15次迭代后的紧凑性、泛化性和特异性。对于BU-

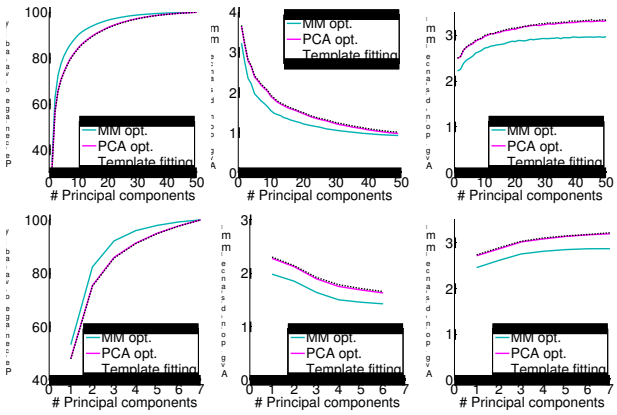


图8. 模板拟合[27]、PCA优化[10]（PCA opt.）与多线性模型优化（MM opt.）在BU-3DFE子集上的比较。左：紧凑性。中：泛化性。右：特异性。上：身份模式。下：表情模式。

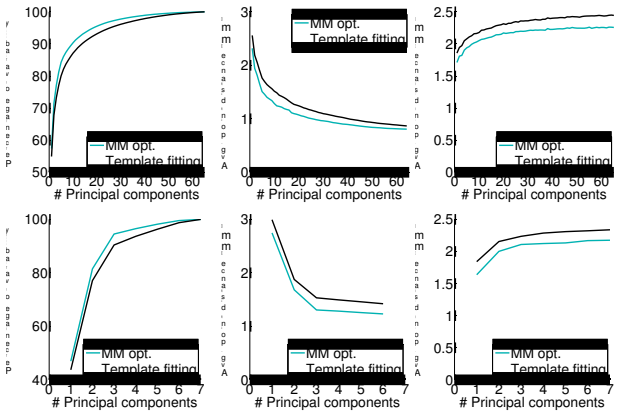


图9：模板拟合[27]与多线性模型优化（MM opt.）在Bosphorus子集上的比较。左：紧致性。中：泛化性。右：特异性。上：身份模式。下：表情模式。

3DFE子集上，平均紧致度提升了3.0%，平均泛化性和特异性分别下降了0.25mm和0.32mm。对于Bosphorus子集，平均紧致度提升了1.7%，平均泛化性和特异性分别下降了0.15mm和0.16mm。

图7直观地比较了BU-3DFE子集中一个对象的模板拟合（红色）与我们的结果（蓝色）。优化前，外部边界的形状存在差异。优化减小了第一和第四种表情在脸颊部位的面部范围，第二种表情在下颌部位，第三种表情在前额部位。第一、二、三种表情在前额处有所扩展。经过15次迭代后，外部边界趋于相似。

为展示我们的方法在大型形状集合上进行优化的能力，我们考虑了Bosphorus数据库的第二个子集，该子集包含39个身份执行

26 action units each, leading to a total of over 1000 shapes. To keep 95% of the data variability after template fitting, a total of 27 components are necessary, while after 15 iterations of our optimization, 20 components suffice. As for the other subsets, generalization and specificity also improve after optimization. To the best of our knowledge, this is the first time a registration optimization based on MDL has been applied to such a large set of shapes.

For all three datasets the model improves significantly during optimization, leading to a more compact model with improved generalization and specificity.

PCA: For brevity, we abbreviate PCA optimization by PCA opt. and our method by MM opt. during the discussion of the comparison. We start by comparing the computational complexity of the two methods. In the supplementary material, we show that one optimization step for PCA opt. has complexity $O(nd_2^2d_3^2)$, while one optimization step of MM opt. has complexity $O(nd_2^2d_3 + nd_2d_3^2)$. For the BU-3DFE subset our non-optimized implementation takes about 16.2h for MM opt. and about 21.5h for PCA opt. for one iteration when executed on a standard PC.

Figure 8 quantitatively compares PCA opt. and MM opt., both after 15 iterations. While MM opt. gives significant improvements, PCA opt. only slightly improves the correspondence. For small subsets PCA opt. gives significant improvements within few iterations. Our experiments suggest that for an increasing number of shape space parameters, an increasing number of iterations is required. Since MM opt. models identity and expression independently, the number of shape space parameters is d_2+d_3, while for PCA opt. the number of shape space parameters is d_2d_3.

Hence, our method gives better improvements after the same number of iterations and is computationally faster than existing linear optimization methods.

5.5. Discussion

Parametrization: Our proposed method optimizes the correspondence by re-parametrizing the shapes guided by the optimization of a multilinear compactness objective function. This re-parametrization requires a continuous representation of the surface for each shape. While any kind of continuous mapping can be used, we establish this by a thin-plate spline. For other continuous mappings, the gradient changes, and therefore depending on the mapping (*e.g.* for mappings without analytical gradient) E must be optimized with a different method.

Data quality: Computing this continuous surface mapping assumes the original face scans to be regularly densely sampled with points that are within the surface of the scan. To get this sampling, any existing template fitting method can be used. For face scans with partial occlusions or strong distortions, template fitting methods fail, since they are unable to estimate the real face surface in these regions. To optimize the registration for scans with strong distortions, we would either need another initialization that gives a reasonable surface estimation within the occluded and noisy regions (*e.g.* Brunton *et al.* [5]), or the optimization of E must be allowed to leave the surface of the disturbed scan guided by the underlying multilinear model.

Computational complexity: While the multilinear correspondence optimization is computationally more efficient than previous linear methods, due to the groupwise objective function, the computational complexity is still high. Our experiments show that only a low number of iterations are necessary to get significant improvements. Note that the registration can be seen as pre-processing that only needs to be done once. The application for larger datasets would require the use of a compute cluster to exploit the full potential of the parallelizability of the method (especially the gradient computation).

Extensions: Our method is generally applicable to other classes of multilinearly distributed data. The geometry of the shapes can contain no or multiple holes as long as the boundaries of the holes are constrained. The regularization E_{REG} prevents fold-overs around these holes. Furthermore, the extension of our method to more modes is straightforward, *e.g.* for faces to associate the fourth mode with viseme or age.

6. Conclusion

We have presented the first method for multilinearly distributed data that jointly improves a given registration and a multilinear model. A continuous representation of each shape allows to optimize the registration with a quasi-Newton method. We have evaluated our method on scans of two databases and have demonstrated that our method is robust to noise in the initial registration. A key advantage of our approach over existing linear MDL methods is its increased computational efficiency, which allows for the first time to apply an approach based on MDL to databases containing over 1000 shapes. We have shown that using the efficient HOSVD method to compute the multilinear model performs similarly when reconstructing unseen face data to more elaborate tensor decompositions. To facilitate experiments for different application scenarios, we make our optimization code and the optimized statistical model available.

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每个动作单元26个，共计超过1000种形状。为在模板拟合后保留95%的数据变异性，共需27个成分，而经过我们优化的15次迭代后，20个成分即已足够。至于其他子集，优化后泛化性和特异性也有所提升。据我们所知，这是首次将基于MDL的配准优化应用于如此大规模的形状集合。

对于所有三个数据集，模型在优化过程中显著提升，最终得到更紧凑的模型，并提高了泛化性和特异性。

PCA：为简洁起见，在比较讨论中，我们将PCA优化简称为PCA opt.，将我们的方法简称为MM opt.。首先比较两种方法的计算复杂度。在补充材料中，我们表明PCA opt.的一步优化复杂度为$O(nd_2^2d_3^2)$，而MM opt.的一步优化复杂度为$O(nd_2^2d_3 + nd_2d_3^2)$。对于BU-3DFE子集，在标准PC上执行时，我们的非优化实现每次迭代MM opt.约需16.2小时，PCA opt.约需21.5小时。

图8在15次迭代后定量比较了PCA优化与MM优化。尽管MM优化带来了显著改进，PCA优化仅略微改善了对应关系。对于小子集，PCA优化在少量迭代内即可取得显著改进。我们的实验表明，随着形状空间参数数量的增加，所需迭代次数也随之增加。由于MM优化分别对身份和表情进行建模，形状空间参数数量为d_2+d_3，而PCA优化的形状空间参数数量为d_2d_3。

因此，我们的方法在相同迭代次数后能取得更好的改进效果，并且在计算上比现有的线性优化方法更快。

5.5. 讨论

参数化：我们提出的方法通过在多线性紧致性目标函数的优化引导下对形状进行重新参数化，从而优化对应关系。这种重新参数化需要每个形状具有连续的曲面表示。虽然可以使用任意类型的连续映射，但我们通过薄板样条来实现这一点。对于其他连续映射，梯度会发生变化，因此根据映射的不同（*e.g*，例如没有解析梯度的映射），E必须使用不同的方法进行优化。

数据质量：计算该连续表面映射的前提是原始人脸扫描具有规则且密集的点采样，且这些点位于扫描表面之内。为获得此类采样，可使用任何现有的模板拟合方法。对于存在部分遮挡或强烈变形的人脸扫描，模板拟合方法会失效，因为它们无法在这些区域估计真实的人脸表面。

为优化强畸变扫描的配准，我们需要另一种初始化方法，使其能在遮挡和噪声区域内给出合理的表面估计（*e.g*. Brunton *et al*. [5]），或者必须允许对 E 的优化在底层多线性模型的引导下偏离受干扰扫描的表面。

计算复杂度：尽管多线性对应优化在计算上比先前的线性方法更高效，但由于分组目标函数，计算复杂度仍然较高。我们的实验表明，仅需少量迭代即可获得显著改进。需要注意的是，配准可视为仅需执行一次的预处理步骤。对于更大数据集的应用，则需要使用计算集群以充分发挥该方法并行化潜力（尤其是梯度计算方面）。

扩展：我们的方法普遍适用于其他类别的多线性分布数据。形状的几何结构可以包含零个或多个孔洞，只要孔洞的边界受到约束即可。正则化E_{REG}可防止这些孔洞周围发生折叠。此外，将我们的方法扩展到更多模态是直接可行的，*e.g*。对于面部，可将第四模态与视素或年龄关联起来。

6. 结论

我们提出了一种针对多线性分布数据的首种方法，该方法能够联合优化给定的配准结果和多线性模型。通过每个形状的连续表示，我们得以使用拟牛顿方法优化配准过程。我们在两个数据库的扫描数据上评估了我们的方法，并证明其对初始配准中的噪声具有鲁棒性。与现有线性MDL方法相比，我们方法的一个关键优势在于计算效率的提升，这使得基于MDL的方法首次能够应用于包含超过1000个形状的数据库。我们展示了使用高效的HOSVD方法计算多线性模型，在重构未见人脸数据时，其性能与更复杂的张量分解方法相当。为便于不同应用场景的实验，我们公开了优化代码及优化后的统计模型。

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References

[1] V. Blanz and T. Vetter. A morphable model for the synthesis of 3D faces. In *SIGGRAPH*, pages 187–194, 1999.

[2] T. Bolkart and S. Wuhrer. 3D faces in motion: Fully automatic registration and statistical analysis. *CVIU*, 131:100–115, 2015.

[3] T. Bolkart and S. Wuhrer. Multilinear mdl for 3D faces, 2015. http://multilinear-mdl.gforge.inria.fr/.

[4] C. Bregler, A. Hertzmann, and H. Biermann. Recovering non-rigid 3D shape from image streams. In *CVPR*, volume 2, pages 690–696, 2000.

[5] A. Brunton, T. Bolkart, and S. Wuhrer. Multilinear wavelets: A statistical shape space for human faces. In *ECCV*, pages 297–312, 2014.

[6] O. Burghard, A. Berner, M. Wand, N. J. Mitra, H. Seidel, and R. Klein. Compact part-based shape spaces for dense correspondences. *CoRR*, abs/1311.7535, 2013.

[7] E. Candès, M. Wakin, and S. Boyd. Enhancing sparsity by reweighted l_1 minimization. *JFAA*, 14(5-6):877–905, 2008.

[8] J.-H. Chen, K. C. Zheng, and L. G. Shapiro. 3D point correspondence by minimum description length in feature space. In *ECCV*, pages 621–634, 2010.

[9] R. Davies, C. Twining, T. Cootes, and C. Taylor. Building 3-d statistical shape models by direct optimization. *MI*, 29(4):961–981, 2010.

[10] R. Davies, C. Twining, and C. Taylor. *Statistical Models of Shape: Optimisation and Evaluation*. Springer, 2008.

[11] I. Dryden and K. Mardia. *Statistical shape analysis*. Wiley, 1998.

[12] L. Eldén and B. Savas. A newton-grassmann method for computing the best multilinear rank-(r_1, r_2, r_3) approximation of a tensor. *SIAM J. Matrix Anal. Appl.*, 31(2):248–271, 2009.

[13] S. T. Gollmer, M. Kirschner, T. M. Buzug, and S. Wesarg. Using image segmentation for evaluating 3D statistical shape models built with groupwise correspondence optimization. *CVIU*, 125(0):283 – 303, 2014.

[14] J. Guo, X. Mei, and K. Tang. Automatic landmark annotation and dense correspondence registration for 3D human facial images. *BMC Bioinf.*, 14(1), 2013.

[15] C. J. Hillar and L.-H. Lim. Most tensor problems are NP-hard. *JACM*, 60(6):45:1–45:39, 2013.

[16] D. Hirshberg, M. Loper, E. Rachlin, and M. Black. Coregistration: Simultaneous alignment and modeling of articulated 3D shape. In *ECCV*, pages 242–255, 2012.

[17] Y. Huang, X. Zhang, Y. Fan, L. Yin, L. Seversky, J. Allen, T. Lei, and W. Dong. Reshaping 3D facial scans for facial appearance modeling and 3D facial expression analysis. *IVC*, 30(10):750 – 761, 2012.

[18] M. Irani. Multi-frame correspondence estimation using subspace constraints. *Int. J. Comput. Vision*, 48(3):173–194, 2002.

[19] L. Kobbelt, S. Campagna, J. Vorsatz, and H.-P. Seidel. Interactive multi-resolution modeling on arbitrary meshes. In *SIGGRAPH*, pages 105–114, 1998.

[20] T. G. Kolda and B. W. Bader. Tensor decompositions and applications. *SIAM Rev.*, 51(3):455–500, 2009.

[21] A. C. Kotcheff and C. J. Taylor. Automatic construction of eigenshape models by direct optimization. *Med. Image Anal.*, 2(4):303 – 314, 1998.

[22] L. D. Lathauwer. *Signal processing based on multilinear algebra*. PhD thesis, K.U. Leuven, Belgium, 1997.

[23] D. Liu and J. Nocedal. On the limited memory method for large scale optimization. *Math. Prog.: Series A and B*, 45(3):503–528, 1989.

[24] I. Mpiperis, S. Malassiotis, and M. G. Strintzis. Bilinear models for 3-D face and facial expression recognition. *IFS*, 3:498–511, 2008.

[25] G. Pan, X. Zhang, Y. Wang, Z. Hu, X. Zheng, and Z. Wu. Establishing point correspondence of 3D faces via sparse facial deformable model. *IP*, 22(11):4170–4181, 2013.

[26] G. Passalis, P. Perakis, T. Theoharis, and I. Kakadiaris. Using facial symmetry to handle pose variations in real-world 3D face recognition. *PAMI*, 33(10):1938–1951, 2011.

[27] A. Salazar, S. Wuhrer, C. Shu, and F. Prieto. Fully automatic expression-invariant face correspondence. *MVAP*, pages 1–21, 2013.

[28] A. Savran, N. Alyuz, H. Dibeklioglu, O. Celiktutan, B. Gökberk, B. Sankur, and L. Akarun. Bosphorus database for 3D face analysis. In *BIOID*, pages 47–56, 2008.

[29] L. Torresani, D. Yang, E. Alexander, and C. Bregler. Tracking and modeling non-rigid objects with rank constraints. In *CVPR*, pages 493–500, 2001.

[30] D. Vlasic, M. Brand, H. Pfister, and J. Popović. Face transfer with multilinear models. *SIGGRAPH*, 24(3):426–433, 2005.

[31] F. Yang, L. Bourdev, J. Wang, E. Shechtman, and D. Metaxas. Facial expression editing in video using a temporally-smooth factorization. In *CVPR*, pages 861–868, 2012.

[32] L. Yin, X. Wei, Y. Sun, J. Wang, and M. J. Rosato. A 3D facial expression database for facial behavior research. In *FG*, pages 211–216, 2006.

参考文献

[1] V. Blanz 和 T. Vetter。用于三维人脸合成的可变形模型。见 *SIGGRAPH*，第 187–194 页，1999 年。[2] T. Bolkart 和 S. Wuhrer。运动中的三维人脸：全自动配准与统计分析。*CVIU*，131:100–115，2015 年。[3] T. Bolkart 和 S. Wuhrer。用于三维人脸的多线性 mdl，2015 年。http://multilinear-mdl.gforge.inria.fr/。[4] C. Bregler、A. Hertzmann 和 H. Biermann。从图像流中恢复非刚性三维形状。见 *CVPR*，第 2 卷，第 690–696 页，2000 年。[5] A. Brunton、T. Bolkart 和 S. Wuhrer。多线性小波：人脸的统计形状空间。见 *ECCV*，第 297–312 页，2014 年。[6] O. Burghard、A. Berner、M. Wand、N. J. Mitra、H. Seidel 和 R. Klein。用于密集对应的紧凑分部形状空间。*CoRR*，abs/1311.7535，2013 年。[7] E. Candès、M. Wakin 和 S. Boyd。通过重新加权的 l_1 最小化增强稀疏性。*JFAA*，14(5-6):877–905，2008 年。[8] J.-H. Chen、K. C. Zheng 和 L. G. Shapiro。特征空间中基于最小描述长度的三维点对应。见 *ECCV*，第 621–634 页，2010 年。[9] R. Davies、C. Twining、T. Cootes 和 C. Taylor。通过直接优化构建三维统计形状模型。*MI*，29(4):961–981，2010 年。[10] R. Davies、C. Twining 和 C. Taylor。*Statistical Models of Shape: Optimisation and Evaluation*。Springer，2008 年。[11] I. Dryden 和 K. Mardia。*Statistical shape analysis*。Wiley，1998 年。[12] L. Eldén 和 B. Savas。一种用于计算张量最佳多线性秩-(r_1, r_2, r_3)逼近的牛顿-格拉斯曼方法。*SIAM J. Matrix Anal. Appl.*，31(2):248–271，2009 年。[13] S. T. Gollmer、M. Kirschner、T. M. Buzug 和 S. Wesarg。使用图像分割评估基于群组对应优化构建的三维统计形状模型。*CVIU*，125(0):283–303，2014 年。[14] J. Guo、X. Mei 和 K. Tang。三维人脸图像的自动地标标注与密集对应配准。*BMC Bioinf.*，14(1)，2013 年。[15] C. J. Hillar 和 L.-H. Lim。大多数张量问题是 NP 难的。*JACM*，60(6):45:1–45:39，2013 年。[16] D. Hirshberg、M. Loper、E. Rachlin 和 M. Black。协同配准：关节三维形状的同时对齐与建模。见 *ECCV*，第 242–255 页，2012 年。[17] Y. Huang、X. Zhang、Y. Fan、L. Yin、L. Seversky、J. Allen、T. Lei 和 W. Dong。重塑三维人脸扫描以实现人脸外观建模和三维表情分析。*IVC*，30(10):750–761，2012 年。[18] M. Irani。利用子空间约束的多帧对应估计。*Int. J. Comput. Vision*，48(3):173–194，2002 年。[19] L. Kobbelt、S. Campagna、J. Vorsatz 和 H.-P. Seidel。任意网格上的交互式多分辨率建模。见 *SIGGRAPH*，第 105–114 页，1998 年。[20] T. G. Kolda 和 B. W. Bader。张量分解及其应用。*SIAM Rev.*，51(3):455–500，2009 年。[21] A. C. Kotcheff 和 C. J. Taylor。通过直接优化自动构建特征形状模型。*Med. Image Anal.*，2(4)：303–314，1998年。[22] L. D. Lathauwer。*Signal processing based on multilinear algebra*。博士论文，比利时鲁汶大学，1997年。[23] D. Liu 和 J. Nocedal。关于大规模优化的有限内存方法。*Math. Prog.: Series A and B*，45(3)：503–528，1989年。[24] I. Mpiperis、S. Malassiotis 和 M. G. Strintzis。用于3D人脸和面部表情识别的双线性模型。*IFS*，3：498–511，2008年。[25] G. Pan、X. Zhang、Y. Wang、Z. Hu、X. Zheng 和 Z. Wu。通过稀疏面部可变形模型建立3D人脸的对应点。*IP*，22(11)：4170–4181，2013年。[26] G. Passalis、P. Perakis、T. Theoharis 和 I. Kakadiaris。利用面部对称性处理真实世界3D人脸识别中的姿态变化。*PAMI*，33(10)：1938–1951，2011年。[27] A. Salazar、S. Wuhrer、C. Shu 和 F. Prieto。全自动表情不变的人脸对应。*MVAP*，页码1–21，2013年。[28] A. Savran、N. Alyuz、H. Dibeklioglu、O. Celiktutan、B. Gökberk、B. Sankur 和 L. Akarun。用于3D人脸分析的Bosphorus数据库。载于*BIOID*，页码47–56，2008年。[29] L. Torresani、D. Yang、E. Alexander 和 C. Bregler。利用秩约束跟踪和建模非刚性物体。载于*CVPR*，页码493–500，2001年。[30] D. Vlasic、M. Brand、H. Pfister 和 J. Popović。多线性模型的人脸迁移。*SIGGRAPH*，24(3)：426–433，2005年。[31] F. Yang、L. Bourdev、J. Wang、E. Shechtman 和 D. Metaxas。使用时间平滑分解的视频面部表情编辑。载于*CVPR*，页码861–868，2012年。[32] L. Yin、X. Wei、Y. Sun、J. Wang 和 M. J. Rosato。用于面部行为研究的3D面部表情数据库。载于*FG*，页码211–216，2006年。